Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(Data Science) Sem-IV Regular/Remedial April-June 2024

U74A1GUI: GUI PROGRAMMING

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions. Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Write a short note on .Net Framework.	06	CO1	BTL5
2	Explain how to write console application with example.	06	CO2	BTL6
3	What is Namespace? Explain it in details.	06	CO1	BTL3
4	Discuss Jagged Array with suitable example.	06	CO3	BTL6
5	Illustrate the use of Exception handling in C Sharp.	06	CO3	BTL5
6	Explain Check Box and Radio Button Control with its properties, methods and attribute.	06	CO4	BTL3
7	Explain Connection with SQL server in detail.	06	CO5	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Define JIT and explain it with it's types in details.	06	CO1	BTL3
2	Write a short note on CLS.	06	CO1	BTL3
3	Explain Design window and Class window.	06	CO2	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Elaborate the use of Abstraction, Interface, and Polymorphism.	06	CO3	BTL5
2	Explain Tree view and List view Control with its properties, methods and attribute.	06	CO4	BTL3
3	Discuss the use of Open File Dialog & Save File Dialog.	06	CO4	BTL3
Q.4	**Answer the following: (Any One)**	**06**		
1	Discuss the Architecture of ADO.Net.	06	CO5	BTL5
2	Explain Data Set object in detail.	06	CO5	BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B. Sc.IT (Data Science) Sem-IV Regular/Remedial April-June 2024

U74A2WD2: WEB DESIGNING-II

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Analyze the various effects available in jQuery.	06	CO1	BTL4
2	What is jQuery? Elaborate the uses and advantages of jQuery.	06	CO1	BTL2/BTL4
3	Analyze the jQuery plugging in details.	06	CO2	BTL4
4	Categorize different background properties of CSS3.	06	CO3	BTL2/BTL4
5	Explain the use of animation with different properties and values in CSS.	06	CO3	BTL4
6	What is media-query? Analyze various device break-points of media query.	06	CO4	BTL2/BTL4
7	Analyze the badges in Bootstrap.	06	CO5	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Analyze any five Event handling in jQuery.	06	CO1	BTL4
2	What is an AJAX? Simplify various technologies involved in AJAX.	06	CO2	BTL2/BTL4
3	Elaborate $.load() and $.ajax() with proper example.	06	CO2	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	What is a CSS? Compare various types of CSS in brief.	06	CO3	BTL2/BTL5
2	What is Viewport? How can we set the viewport?	06	CO4	BTL2/BTL4
3	Demonstrate the different between Adaptive v/s Responsive Web Design.	06	CO4	BTL3

Q.4	**Answer the following: (Any One)**	**06**		
1	Evaluate different types of classes to create different styles of button in Bootstrap.	06	CO5	BTL5
2	Explain various classes of Bootstrap to create different styles of table.	06	CO5	BTL6

**** End of the Paper ****

BTL5

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DS) Sem-IV Reg/Rem April-June 2024

U74A3DV: DATA VISUALIZATION

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Explain the application of data visualization.	06	CO1	BTL2
2	Analyze the varied data types supported in Tableau for visualization, emphasizing their uses and comparisons.	06	CO2	BTL4
3	Evaluate data blending and its applications, alongside outlining steps for executing data blending.	06	CO3	BTL5
4	Justify the significance of table calculated fields and elucidate their necessity in creating calculation fields.	06	CO4	BTL5
5	Classify and elaborate on the diverse chart types utilized for efficient data visualization.	06	CO5	BTL4
6	Elaborate comprehensively on the concept of LOD (Level of Detail) in data analysis.	06	CO4	BTL2
7	Compare and contrast the methodologies of data joining and data blending, highlighting their respective advantages and limitations.	06	CO3	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Identify and compare various tools utilized in data visualization, providing concise explanations of each.	06	CO1	BTL4
2	Define Tableau and delve into its features, advantages, and drawbacks for comprehensive understanding.	06	CO2	BTL5
3	Explore the essential elements contributing to successful data visualization, offering a thorough discussion on their significance.	06	CO2	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Compare group, bin, and set functionalities in Tableau.	06	CO3	BTL3
2	Elucidate the significance and practical applications of filters in Tableau, accompanied by an illustrative example demonstrating how filters facilitate the analysis of	06	CO4	BTL3

	targeted data subsets within a dataset.			
3	Clarify the concept of a table calculation filter and its role in data manipulation and analysis within Tableau.	06	CO4	BT
Q.4	**Answer the following: (Any One)**	**06**		
1	Elaborate on the characteristics and functions of Bar Charts and Scatter Plots, providing comprehensive details.	06	CO5	BTL4
2	Provide a detailed explanation of Waterfall Charts and Histograms, highlighting their respective features and applications.	06	CO5	BTL4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DS) Sem-IV Regular/Remedial April-June 2024

U74A4SE: SOFTWARE ENGINEERING

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively.Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern:BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	What is a process model? Evaluate any one process model.	06	CO2	BTL5
2	Discuss various software applications.	06	CO1	BTL3
3	Elaborate the Business Process Engineering Approach in requirement engineering.	06	CO3	BTL4
4	Write a short note on Testing approaches.	06	CO5	BTL4
5	Could you elaborate Umbrella Activities in Software engineering?	06	CO1	BTL4
6	Explain class based requirement modelling strategies.	06	CO4	BTL4
7	What is software testing? What are the **roles** of tester software engineering?	06	CO5	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	What do understand by RAD model? Discuss the diagram of the RAD model in software engineering.	06	CO2	BTL5
2	Describe the project task for planning.	06	CO2	BTL3
3	Illustrate Maturity Levels in software engineering	06	CO1	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Classify the requirement engineering tasks.	06	CO3	BTL3
2	What is Analysis model? Discuss elements of the analysis model.	06	CO4	BTL4
3	Discuss the Concepts of data modelling with give an example.	06	CO4	BTL6
Q.4	**Answer the following: (Any One)**	**06**		
1	Distinguish between alpha testing and beta testing.	06	CO5	BTL3
2	Explain State different between Functional and Non-Functional Testing.	06	CO5	BTL4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT(DS) Sem-IV Regular April-June 2024

Subject: U74A5OR: OPERATION RESEARCH

Total: 60 Marks

Time: 3 Hours

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

		Marks	CO	Bloom's Taxonomy
Q-1	**Answer the following. (Any 5)**	30		
1	Solve following linear programming problem by graphical method.	06	CO1	BTL5

$$Max\ Z = 6x_1 - 4x_2$$

Subject to the constraints,

$$2x_1 + 4x_2 \leq 4,$$
$$4x_1 + 8x_2 \geq 16,$$
$$x_1, x_2 \geq 0.$$

2	Solve following linear programming problem by simplex method	06	CO1	BTL4

$$Max\ Z = 20x_1 + 6x_2 + 8x_3$$

Subject to the constraints,

$$8x_1 + 2x_2 + 3x_3 \leq 250,$$
$$4x_1 + 3x_2 \leq 150,$$
$$2x_1 + x_3 \leq 50,$$
$$x_1, x_2, x_3 \geq 0.$$

3	Obtain the initial basic feasible solution of transportation problem by NWCM.	06	CO2	BTL3

Distribution / Origin	D_1	D_2	D_3	Supply
S_1	2	7	4	5
S_2	3	3	1	8
S_3	5	4	7	7
S_4	1	6	2	14
Demand	7	9	18	

4 Find the optimal solution of the transportation problem. 06 CO2 BTL5

origin	Destination				supply
	D_1	D_2	D_3	D_4	
O_1	3	1	7	4	250
O_2	2	6	5	9	350
O_3	8	3	3	2	400
Demand	200	300	350	150	

5 Four professors are each capable of teaching any one of the four different subject. Class preparation time (in hrs) for different topics varies from professor to professor and is given in the following table. Each professor should be assigned only one subject find the schedule so as to minimize the total subject preparation time for all subjects/professors. 06 CO3 BTL4

Professors	Subjects			
	S_1	S_2	S_3	S_4
P_1	2	10	9	7
P_2	15	4	14	8
P_3	13	14	16	11
P_4	3	15	13	8

6 Consider the processing time (in minute) of 5 jobs each of which must passing through the two machines M_1 *and* M_2 in orderM_1M_2. Find sequence for job that minimize total elapsed time and idle time for M_1 and M_2. 06 CO4 BTL6

Job	J_1	J_2	J_3	J_4	J_5
M_1	5	1	9	3	10
M_2	2	6	7	8	4

7 Solve following linear programming problem. 06 CO1 BTL5

$$Max\ Z = 15x_1 + 10x_2$$

Subject to the constraints,

$$4x_1 + 6x_2 \leq 360,$$
$$3x_1 \leq 180,$$
$$5x_2 \leq 200,$$
$$x_1, x_2 \geq 0.$$

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy
1	Solve following linear programming problem by graphical method	06	CO1	BTL4

$$Max\ Z = 3x_1 + 2x_2$$

Subject to the constraints,

$$x_1 - x_2 \geq 1$$
$$x_1 + x_2 \geq 3,$$
$$x_1, x_2 \geq 0.$$

2 Obtain the initial basic feasible solution of above transportation problem by LCM. — 06 — CO2 — BTL3

Distribution / Origin	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

3 Find the optimal solution of the transportation problem. — 06 — CO2 — BTL5

origin	Destination				supply
	D_1	D_2	D_3	D_4	
O_1	2	2	2	1	3
O_2	10	8	5	4	7
O_3	7	6	6	8	5
Demand	4	3	4	4	

Q.3	Answer the following: (Any Two)	12		
1	Solve the following Assignment problem by Hungarian method.	06	CO3	BTL5

	J_1	J_2	J_3
P_1	10	20	30
P_2	20	10	40
P_3	50	30	20

2 A computer center has three expert programmers. The center wants three application programmers to be developed estimate the computer time in minutes required by experts for application program. — 06 — CO3 — BTL6

		Program		
		A	B	C
Programmers	1	120	100	80
	2	80	90	110
	3	110	140	120

3 Solve the following Assignment problem by Hungarian method. — 06 — CO3 — BTL5

Jobs / Men	J_1	J_2	J_3	J_4
M_1	12	30	21	15
M_2	18	33	9	31
M_3	44	25	24	21
M_4	23	30	28	14

06

Q.4 Answer the following: (Any One) 06 CO4 BTL6

1 A book binder has one printing press, one binding machine and manufactured 7 different books. The time required for performing printing and binding operations for different books are shown below then find sequence that minimize total time.

Book	1	2	3	4	5	6	7
Printing time (hrs)	20	90	80	20	120	15	65
Binding time (hrs)	25	60	75	30	90	35	50

2 Use Graphical Method to minimize the time needed to process the following jobs on the machines as shown in the below table. Find total elapsed time to complete both jobs. 06 CO4 BTL6

		Machine				
Job 1	Sequence	*A*	*B*	*C*	*D*	*E*
	Time(hrs)	3	4	2	6	2

		Machine				
Job 2	Sequence	*B*	*C*	*A*	*D*	*E*
	Time(hrs)	5	4	3	2	6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY
B. Sc.IT (DS) Sem-IV Regular April-June 2024
U74B6SQA: SOFTWARE QUALITY ASSURANCE

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
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5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any five)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	What is software testing, and why is it essential in software development?	06	CO1	BTL1
2	Analyze the verification and validation in software testing.	06	CO1	BTL4
3	Compare alpha and beta testing in software testing.	06	CO2	BTL3
4	Describe performance testing and its objectives.	06	CO2	BTL2
5	Explain the Types of a test strategy.	06	CO3	BTL4
6	Illustrate the process of test case design and its importance.	06	CO3	BTL5
7	Outline the defect life cycle and the stages involved.	06	CO4	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Compare static testing with dynamic testing.	06	CO1	BTL3
2	Differentiate white box testing and Black box testing	06	CO2	BTL3
3	Describe field-level validation and its role in data validation.	06	CO2	BTL2
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain Test Strategy Documents.	06	CO3	BTL1
2	Analyze the concept of database testing and its Significance.	06	CO3	BTL4
3	Explain the concept of test management and its responsibilities.	06	CO4	BTL2
Q.4	**Answer the following: (Any One)**	**06**		
1	What is a defect, and how does it impact software Quality	06	CO4	BTL3
2	Discuss the origins of defects in software development.	06	CO4	BTL4

**** End of the Paper ****